IoT Predictive Maintenance Platform

Summary:

Develop an AI-powered platform that analyzes IoT sensor data from industrial equipment to predict failures, schedule maintenance, and optimize equipment performance.

Problem Statement:

Industrial IoT deployments generate vast amounts of sensor data that can predict equipment failures before they occur. The challenge is to process real-time sensor data, identify patterns indicating potential failures, and recommend optimal maintenance schedules. The system must support multiple equipment types, provide cost-benefit analysis for maintenance decisions, and integrate with existing maintenance management systems.

Proposed Steps:

• Design a scalable data ingestion pipeline for IoT sensor streams.
• Implement anomaly detection algorithms for sensor types (vibration, temperature, pressure).
• Create predictive models for failure modes using time series analysis.
• Build a maintenance scheduling optimization engine considering costs and priorities.
• Develop a monitoring dashboard with real-time equipment health status.
• Include mobile alerts and work order integration for maintenance teams.

Suggested Data Requirements:

• Time-series sensor data from multiple equipment types (temperature, vibration, pressure). Synthetic data can be generated if needed.
• Equipment maintenance history and failure records.
• Spare parts inventory and cost information.
• Equipment specifications and operating parameters.

Themes:

AI in Service Lines, Classical AI/ML/DL for prediction